Phase 4: Statistics & Business Applications (Month 3)

Apply statistical concepts to practical business situations using case studies from aviation, sports management, agriculture, entertainment, and energy sectors.

Section 1: Descriptive vs. Inferential Statistics

Exercise 1: Airline Passenger Analysis

An airline recorded the number of passengers (in hundreds) traveling on 12 domestic flights.

Flight Passengers (Hundreds): A-18, B-22, C-20, D-25, E-19, F-24, G-21, H-23, I-26, J-20, K-27, L-22

1. Calculate mean, median, mode, and standard deviation.
2. Interpret the standard deviation for passenger variation.
3. Identify which statistics are descriptive statistics.
4. Explain whether forecasting future passenger demand is inferential statistics.

Section 2: Mean, Median, Mode & Standard Deviation

Exercise 2: Professional Athlete Earnings Analysis

Annual Earnings ($ Million): 1.5; 1.8; 2.0; 2.2; 2.4; 2.5; 2.7; 15.0

1. Calculate mean, median, range, and standard deviation.
2. Explain why the mean is significantly different from the median.
3. Which measure better represents the earnings of most athletes?
4. Analyze the impact of removing the highest-paid athlete.
5. Explain the effect of outliers on salary analysis.

Section 3: Probability & Business Risk

Exercise 3: Crop Failure Risk Assessment

Probability of Crop Failure:
Low Risk: 4%, Medium Risk: 9%, High Risk: 18%

Farms Insured:
300 Low Risk, 450 Medium Risk, 250 High Risk

1. Calculate the expected number of crop failures for each category.

2. Calculate the total expected crop failures.

3. Identify the highest-risk farming category.

4. Recommend where insurance companies should focus risk reduction efforts.

5. Explain why probability analysis is important in agriculture insurance.

Section 4: Hypothesis Testing

Exercise 4: Movie Theater Customer Satisfaction

A cinema chain claims average customer satisfaction is 90 points out of 100.
Sample mean satisfaction score = 84; Standard deviation = 11.

1. Define H■ and H■.

2. Determine whether management should investigate customer service quality.

3. Explain Type I and Type II errors in this scenario.

4. Discuss the business impact of incorrect conclusions.

5. Explain the importance of hypothesis testing in customer experience management.

Section 5: A/B Testing for Marketing Decisions

Exercise 5: Music Streaming App Promotion

Promotion A: 7,500 users viewed, 525 premium subscriptions
Promotion B: 7,500 users viewed, 690 premium subscriptions

1. Calculate conversion rates for both promotions.

2. Identify the better-performing promotion.

3. Calculate the percentage improvement.

4. Explain how A/B testing supports business decisions.

5. Identify risks reduced through testing.

6. Suggest additional metrics to evaluate campaign performance.

Section 6: Data-Driven Decision Making

Exercise 6: Solar Power Plant Expansion Decision

Region A: Revenue 620, Growth 5%, Competitors 10
Region B: Revenue 540, Growth 11%, Competitors 4
Region C: Revenue 700, Growth 3%, Competitors 16

1. Which region currently generates the highest revenue?

2. Which region has the strongest long-term growth potential?

3. What additional data should be collected before expansion?

4. Explain the importance of data-driven decisions in energy investments.

5. Recommend a region for expansion with justification.
